Project Report: Heart Disease Data Analysis

College: KIET Group of Institutions

Branch: CSE-4E

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Team Members:

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1. Project Objective

The primary goal of this project is to analyze clinical parameters like age, sex, and cholesterol levels to identify patterns associated with heart disease. We used **SQL** for data preprocessing and **Tableau** for creating interactive visual dashboards.

2. Technical Stack

- **Database:** SQL Workspace (Data Cleaning & Filtering)
- **Visualization:** Tableau Desktop (Trend Analysis)
- **Version Control:** GitHub (Project Documentation)
- **Dataset:** `heart_final.csv`

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3. Methodology

Step 1: Data Cleaning (SQL)

In the SQL workspace, we executed queries to ensure data integrity. We filtered patients based on age brackets and removed any null values.

> *Key Query Example:* `SELECT * FROM heart_data WHERE age > 40 AND chol > 200;`

Step 2: Data Visualization (Tableau)

We developed three specialized sheets to visualize the findings:

1. **Age Distribution:** Identified that the 50-60 age group is at the highest risk.
2. **Cholesterol by Gender:** Analyzed the average cholesterol levels across male and female patients.
3. **Risk Ratio (Pie Chart):** Provided a clear percentage of the population currently categorized under 'Heart Disease Risk'.

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4. Final Insights

- Patients in the **late 50s** show a higher frequency of cardiovascular issues.
- **Cholesterol levels** are a significant clinical indicator, with distinct averages between genders.
- Our analysis helps healthcare providers prioritize high-risk demographics for early screening.

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5. Conclusion

This project demonstrates the power of combining SQL data processing with Tableau's visual analytics to derive actionable health insights. All deliverables (SQL Code, Dataset, and Report) are hosted on our GitHub repository.